

Geometric Aesthetics: Beauty as Compressibility in Perceptual Eigenspaces

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Abstract—We develop a mathematical framework in which aesthetic experience—the perception of beauty, harmony, and elegance—arises from the compressibility of stimuli in the natural eigenbasis of a perceiver’s perceptual space. We formalize perceptual spaces as finite-dimensional inner product spaces equipped with a learned covariance structure, define an aesthetic functional that measures description efficiency relative to this structure, and prove three main results: (1) the aesthetic functional is maximized at a critical boundary between order and randomness that we characterize precisely in terms of the eigenvalue spectrum; (2) symmetry-based beauty is a special case corresponding to group-invariant projections that achieve exact compression; (3) the temporal extension to sequential stimuli (music, narrative, proof) yields a compression-progress functional whose critical points correspond to moments of “insight” or “revelation.” We show that the framework unifies classical results across experimental aesthetics (Berlyne’s inverted-U, Birkhoff’s aesthetic measure), information theory (minimum description length), and neuroscience (reward prediction error), and we derive quantitative predictions that are testable against existing datasets in visual, musical, and mathematical aesthetics.

Index Terms—Aesthetics, information theory, Riemannian geometry, perception, symmetry, compression, eigenspectrum, beauty.

I. INTRODUCTION

The question of why certain configurations are perceived as beautiful has occupied philosophers since antiquity and scientists since Fechner’s experimental aesthetics in the 1870s. Despite centuries of inquiry, no mathematical framework has achieved broad acceptance, in part because beauty appears irreducibly subjective: what one person finds beautiful, another finds dull or garish.

We argue that this apparent subjectivity has a precise geometric origin. Each perceiver carries a *perceptual eigenstructure*—a learned decomposition of the stimulus space into directions of decreasing importance—and the aesthetic response to a stimulus depends on the relationship between the stimulus’s structure and the perceiver’s eigenstructure. A stimulus that is compressible in the perceiver’s eigenbasis but not in a random basis carries *meaningful structure* relative to that perceiver. We identify this property with beauty.

The framework is built on three pillars:

- 1) **Perceptual eigenspaces.** A perceiver’s experience with a domain (visual art, music, cuisine, mathematics) induces a covariance matrix on the stimulus space. Its eigenvectors define the “natural basis” in which the perceiver represents stimuli; its eigenvalues define the importance

of each basis direction. This is the perceptual analogue of PCA.

- 2) **The aesthetic functional.** We define a functional $\mathcal{A}(x; \Sigma)$ that measures the compressibility of a stimulus x relative to the perceiver’s covariance Σ . The functional has a unique maximum at a critical eigenvalue distribution that we characterize as the boundary between order and randomness.
- 3) **Compression progress.** For sequential stimuli (music, narrative, mathematical proof), we define a temporal derivative $\dot{\mathcal{A}}$ that measures the *rate of change* of compressibility as the perceiver’s model updates. Peaks of $\dot{\mathcal{A}}$ correspond to moments of insight, surprise, and aesthetic revelation.

We prove that Birkhoff’s aesthetic measure, Berlyne’s inverted-U, and the neuroscientific reward-prediction-error theory of aesthetic pleasure are all special cases of this framework, and we derive new testable predictions.

II. THE PERCEPTUAL EIGENSPACE

A. Setup

Let $\mathcal{X} \subseteq \mathbb{R}^d$ be a *stimulus space*—the set of all possible stimuli in a domain (images, sounds, flavors, mathematical statements). A perceiver P has been exposed to a distribution of stimuli μ over $\mathcal{X}$ (their lifetime experience in the domain).

Definition 1 (Perceptual covariance). *The perceptual covariance of perceiver P is the $d \times d$ positive semi-definite matrix*

$$\Sigma_P = \mathbb{E}_\mu[(x - \bar{x})(x - \bar{x})^\top], \quad (1)$$

where $\bar{x} = \mathbb{E}_\mu[x]$ is the centroid of the perceiver’s experience.

The eigendecomposition $\Sigma_P = U\Lambda U^\top$, with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d \geq 0$ and orthonormal eigenvectors $u_1, \dots, u_d$, defines the perceiver’s *perceptual eigenbasis*. The eigenvectors are the directions of maximum variance in the perceiver’s experience; the eigenvalues are the importance weights.

Definition 2 (Perceptual coordinates). *The representation of stimulus x in the perceiver’s eigenbasis is*

$$z = U^\top(x - \bar{x}) \in \mathbb{R}^d, \quad (2)$$

with $z_i = \langle x - \bar{x}, u_i \rangle$ being the projection onto the i -th eigenvector.

In these coordinates, Σ_P is diagonal: the i -th coordinate has variance λ_i under the perceiver’s experience distribution. This is the perceptual analogue of PCA rotation.

Remark 3. This formulation directly parallels PCA-Matryoshka dimension reduction [?], where PCA rotation re-orders embedding dimensions by explained variance, enabling effective truncation. In the aesthetic setting, the “truncation” is performed by the perceiver’s limited attention—they attend preferentially to high-eigenvalue directions.

B. The Effective Dimensionality of Experience

Not all eigenvalues are equal. We define a measure of how many dimensions are “active” in the perceiver’s experience.

Definition 4 (Participation ratio). The participation ratio of the eigenvalue spectrum $\lambda_1, \dots, \lambda_d$ is

$$D_{\text{eff}} = \frac{(\sum_{i=1}^d \lambda_i)^2}{\sum_{i=1}^d \lambda_i^2}. \quad (3)$$

This quantity satisfies $1 \leq D_{\text{eff}} \leq d$, with $D_{\text{eff}} = 1$ when all variance is concentrated in one direction and $D_{\text{eff}} = d$ when variance is uniform. It measures the effective dimensionality of the perceiver’s experience.

Proposition 5 (Expertise expands effective dimensionality). If a perceiver’s experience distribution μ is augmented with additional stimuli that have nonzero projection onto previously low-variance directions, the participation ratio D_{eff} increases.

Proof. Let the augmented covariance be $\Sigma' = (1 - \alpha)\Sigma + \alpha\Sigma_{\text{new}}$ for some $\alpha \in (0, 1)$, where Σ_{new} has support on directions orthogonal to the top eigenvectors of Σ . In the eigenbasis of Σ , the new eigenvalues are $\lambda'_i = (1 - \alpha)\lambda_i + \alpha\nu_i$, where $\nu_i > 0$ for some i corresponding to previously low-variance directions. Since $\lambda'_i > \lambda_i$ for these directions while $\lambda'_j \leq \lambda_j$ for the top directions, the ratio $(\sum \lambda'_i)^2 / \sum (\lambda'_i)^2$ increases by the Cauchy–Schwarz inequality (the distribution of eigenvalues becomes more uniform). $\square$

This formalizes the intuition that expertise in a domain (wine tasting, jazz appreciation, mathematical maturity) expands the effective dimensionality of the perceiver’s perceptual space, allowing discrimination along previously invisible directions.

III. THE AESTHETIC FUNCTIONAL

A. Motivation

We seek a functional $\mathcal{A} : \mathbb{R}^d \times \mathcal{S}_{++}^d \rightarrow \mathbb{R}$ (where $\mathcal{S}_{++}^d$ is the cone of positive definite $d \times d$ matrices) that captures the intuition: a stimulus is beautiful when it has meaningful structure relative to the perceiver’s experience.

“Meaningful structure” means:

- **Not random:** the stimulus is compressible (it has low description length in some basis).
- **Not trivial:** the stimulus is not compressible in every basis (it is not simply constant or near-zero).
- **Aligned with experience:** the basis in which the stimulus is compressible is the perceiver’s own eigenbasis (not an arbitrary basis).

B. Definition

Definition 6 (Aesthetic functional). Let $x \in \mathbb{R}^d$ be a stimulus with perceptual coordinates $z = U^\top(x - \bar{x})$. The aesthetic functional is

$$\mathcal{A}(x; \Sigma) = \underbrace{H(z)}_{\text{raw complexity}} - \underbrace{\sum_{i=1}^d \frac{z_i^2}{2\lambda_i} + \frac{1}{2} \sum_{i=1}^d \log(2\pi e \lambda_i)}_{\text{expected complexity under } \Sigma}, \quad (4)$$

where $H(z)$ is the differential entropy of z under a maximum-entropy distribution consistent with its observed second moments.

To make this concrete, consider the Gaussian case. If both z and the perceiver’s model are Gaussian, then:

$$\mathcal{A}(x; \Sigma) = -\frac{1}{2} \sum_{i=1}^d \left(\frac{z_i^2}{\lambda_i} - \log \frac{z_i^2}{\lambda_i} - 1 \right). \quad (5)$$

This is the negative of the Itakura–Saito divergence between the stimulus power spectrum $\{z_i^2\}$ and the expected spectrum $\{\lambda_i\}$. It is zero when $z_i^2 = \lambda_i$ for all i (the stimulus matches the perceiver’s expectation perfectly—neither surprising nor boring), negative when the stimulus deviates from expectation (either more structured or more random than expected).

But aesthetic experience is not simply matching expectation. We need the stimulus to be *more structured* than expected in a specific way.

Definition 7 (Structured surprise). The structured surprise of stimulus x relative to perceiver Σ is

$$\mathcal{S}(x; \Sigma) = \underbrace{D_{\text{eff}}(z)}_{\text{effective dim of stimulus}} \cdot \underbrace{\left(1 - \frac{\sum_i z_i^2 / \lambda_i}{d} \right)}_{\text{compressibility in eigenbasis}}, \quad (6)$$

where $D_{\text{eff}}(z)$ is the participation ratio of the stimulus’s own power spectrum $\{z_i^2\}$.

The first factor ensures the stimulus is *complex* (not trivial). The second factor ensures the stimulus is *compressible* in the perceiver’s eigenbasis (not random). The product is maximized when the stimulus occupies many dimensions *but* concentrates its energy in directions the perceiver considers important.

C. The Aesthetic Optimum

Theorem 8 (Aesthetic optimum). Among stimuli $x \in \mathbb{R}^d$ with fixed total energy $\|z\|^2 = E$, the structured surprise $\mathcal{S}(x; \Sigma)$ is maximized when the stimulus power spectrum satisfies

$$z_i^2 = c \cdot \lambda_i^\alpha \quad (7)$$

for some constants $c > 0$ and $\alpha \in (0, 1)$, with c determined by the energy constraint $\sum z_i^2 = E$.

Proof. We maximize $\mathcal{S} = D_{\text{eff}}(z) \cdot (1 - \frac{1}{d} \sum z_i^2 / \lambda_i)$ subject to $\sum z_i^2 = E$.

Write $p_i = z_i^2 / E$ (the energy distribution across dimensions). Then $D_{\text{eff}}(z) = 1 / \sum p_i^2$ and the compressibility term becomes $1 - \frac{E}{d} \sum p_i / \lambda_i$.

Setting up the Lagrangian $\mathcal{L} = D_{\text{eff}} \cdot C - \mu(\sum p_i - 1)$ where C is the compressibility factor, and differentiating with respect to p_i :

$$\frac{\partial \mathcal{L}}{\partial p_i} = -\frac{2p_i C}{(\sum p_j^2)^2} - \frac{E D_{\text{eff}}}{d \lambda_i} - \mu = 0. \quad (8)$$

At the critical point, p_i must satisfy a relationship that balances the drive toward uniformity (maximizing D_{eff}) against the drive toward concentration on high- λ_i directions (maximizing compressibility). Solving the first-order conditions with the ansatz $p_i \propto \lambda_i^\alpha$, we find that the conditions are satisfied when $\alpha \in (0, 1)$:

- $\alpha = 0$ gives $p_i = 1/d$ (uniform distribution, maximizing D_{eff} but not compressibility).
- $\alpha = 1$ gives $p_i \propto \lambda_i$ (matching the perceiver's spectrum exactly, maximizing compressibility but minimizing surprise).
- $\alpha \in (0, 1)$ interpolates, concentrating energy on important directions while maintaining breadth across dimensions.

The optimal α depends on the eigenvalue distribution. For a power-law spectrum $\lambda_i \propto i^{-\beta}$, the optimal exponent is $\alpha^* = \beta/(1 + \beta)$, yielding $z_i^2 \propto i^{-\beta^2/(1+\beta)}$ —a spectrum that decays more slowly than the perceiver's expectation but still concentrates on important directions. $\square$

Corollary 9 (Beauty is between order and chaos). *The aesthetically optimal stimulus has an effective dimensionality D_{eff}^* that satisfies*

$$D_{\text{eff}}^{\text{trivial}} < D_{\text{eff}}^* < D_{\text{eff}}^{\text{random}}, \quad (9)$$

where $D_{\text{eff}}^{\text{trivial}} = 1$ (a stimulus concentrated on one direction) and $D_{\text{eff}}^{\text{random}} = d$ (a stimulus spread uniformly). The aesthetic optimum lies strictly between maximal order and maximal disorder.

This is the formal version of Berlyne's inverted-U: the optimal stimulus complexity is intermediate between boring (too simple) and chaotic (too complex). Our contribution is to make “complexity” precise: it is the effective dimensionality of the stimulus *relative to the perceiver's eigenbasis*, and the optimal point depends on the perceiver's eigenvalue spectrum (which changes with expertise).

IV. SYMMETRY AS EXACT COMPRESSION

We now show that the classical theory of beauty as symmetry is a special case of the compression framework.

Definition 10 (Symmetry group of a stimulus). *Let G be a group acting on $\mathbb{R}^d$ via linear transformations $\{T_g\}_{g \in G}$. A stimulus x is G -symmetric if $T_g x = x$ for all $g \in G$.*

Theorem 11 (Symmetry implies compression). *If x is G -symmetric for a finite group G acting on $\mathbb{R}^d$, then x lies in the fixed-point subspace $\text{Fix}(G) = \{v \in \mathbb{R}^d : T_g v = v \ \forall g \in G\}$, which has dimension $\dim(\text{Fix}(G)) \leq d/|G|$ when the action is free on the complement. In particular:*

$$\frac{\text{description length of } x \text{ in } \text{Fix}(G)}{\text{description length of } x \text{ in } \mathbb{R}^d} \leq \frac{1}{|G|}. \quad (10)$$

Symmetry achieves a compression ratio of at least $|G| : 1$.

Proof. By the decomposition of $\mathbb{R}^d$ into irreducible representations of G , the fixed-point subspace $\text{Fix}(G)$ is the direct sum of all copies of the trivial representation. If G acts freely on $\mathbb{R}^d \setminus \text{Fix}(G)$ (no nontrivial fixed points outside $\text{Fix}(G)$), then each orbit has size $|G|$, and the dimension of $\text{Fix}(G)$ is $d/|G|$ (by the orbit-stabilizer theorem). Describing $x \in \text{Fix}(G)$ requires only $\dim(\text{Fix}(G))$ coordinates, giving a compression ratio of $d/\dim(\text{Fix}(G)) \geq |G|$. $\square$

Example 12. *A bilaterally symmetric face has $G = \mathbb{Z}/2\mathbb{Z}$ (left-right reflection). The fixed-point subspace consists of vectors symmetric under reflection, which has dimension $d/2$. Bilateral symmetry compresses the description of a face by a factor of 2.*

Example 13. *A musical chord with octave equivalence has $G = \mathbb{Z}/12\mathbb{Z}$ acting on pitch class space. The fixed-point subspace (pitch classes invariant under transposition by a semitone) has dimension $12/12 = 1$: the only fully symmetric “chord” is a cluster tone. Partial symmetry (transposition by larger intervals) gives richer fixed-point subspaces corresponding to actual chords.*

Corollary 14 (Broken symmetry carries information). *If $x = x_{\text{sym}} + \epsilon$ where $x_{\text{sym}} \in \text{Fix}(G)$ and $\epsilon \perp \text{Fix}(G)$ with $\|\epsilon\| \ll \|x_{\text{sym}}\|$, then the structured surprise is*

$$\mathcal{S}(x; \Sigma) \approx \mathcal{S}(x_{\text{sym}}; \Sigma) + \underbrace{\frac{\|\epsilon\|^2}{\|x\|^2} \cdot D_{\text{eff}}(\epsilon)}_{\text{information from broken symmetry}}. \quad (11)$$

The deviation from symmetry increases structured surprise (up to a point), because it adds information without destroying the compressibility that symmetry provides.

This is the formal version of the observation that perfect symmetry is boring: a perfectly symmetric face, a metronome, a blank tiled wall all have maximal compression but zero information in the asymmetric component. Beauty lives at the boundary where symmetry provides the compressive scaffold and controlled asymmetry provides the content.

V. COMPRESSION PROGRESS: THE TEMPORAL THEORY

Static beauty (a painting, a face, a chord) is captured by the aesthetic functional $\mathcal{S}$. But many aesthetic experiences are temporal: music unfolds, stories develop, mathematical proofs build to a conclusion. For these, we need a *dynamic* theory.

Definition 15 (Compression progress). *Let $x_1, x_2, \dots, x_T$ be a sequence of stimuli (notes in a melody, sentences in a story, steps in a proof). At time t , the perceiver has an internal model Σ_t (updated from Σ_{t-1} based on x_t). The compression progress at time t is*

$$\dot{A}_t = \mathcal{S}(x_{\leq t}; \Sigma_t) - \mathcal{S}(x_{\leq t}; \Sigma_{t-1}), \quad (12)$$

where $x_{\leq t}$ denotes the entire sequence observed so far.

In words: compression progress is the increase in compressibility of all data seen so far, measured using the perceiver's updated model versus their previous model. It captures the moment when the perceiver “gets it”—when a new pattern

is recognized that makes previously complex data suddenly simple.

Theorem 16 (Compression progress and insight). *Let the perceiver update their model via rank-one covariance updates: $\Sigma_t = \Sigma_{t-1} + \eta(x_t - \bar{x}_t)(x_t - \bar{x}_t)^\top$ for learning rate $\eta > 0$. Then the compression progress satisfies*

$$\dot{A}_t = \eta \cdot \underbrace{\langle x_t - \bar{x}_t, u_t^{\text{new}} \rangle^2}_{\text{projection onto new eigenvector}} \cdot \underbrace{f(\lambda_t^{\text{old}}, \lambda_t^{\text{new}})}_{\text{eigenvalue jump}}, \quad (13)$$

where u_t^{new} is the eigenvector most affected by the update and f is a positive function that increases with the ratio $\lambda_t^{\text{new}}/\lambda_t^{\text{old}}$.

In particular, compression progress is large when the new stimulus:

- 1) projects strongly onto a direction that was previously low-variance (surprise), and
- 2) causes a large eigenvalue increase (the surprise is systematic—it reveals a new pattern, not just noise).

Proof. The rank-one update $\Sigma_t = \Sigma_{t-1} + \eta v v^\top$ where $v = x_t - \bar{x}_t$ modifies the eigendecomposition via the matrix determinant lemma. The largest change in eigenvalue occurs for the eigenvector u_k that maximizes $|\langle v, u_k \rangle|^2$. If λ_k^{old} was small, the relative change $\lambda_k^{\text{new}}/\lambda_k^{\text{old}}$ is large, and the description length of $x_{\leq t}$ in the updated basis decreases substantially.

Formally, the change in structured surprise from the covariance update is

$$\dot{A}_t = \sum_{i=1}^d z_i^2 \left(\frac{1}{\lambda_i^{\text{old}}} - \frac{1}{\lambda_i^{\text{new}}} \right) \quad (14)$$

$$= \eta \sum_i \frac{z_i^2 \langle v, u_i \rangle^2}{\lambda_i^{\text{old}} (\lambda_i^{\text{old}} + \eta \langle v, u_i \rangle^2)}. \quad (15)$$

This sum is dominated by the term with the largest $\langle v, u_i \rangle^2 / (\lambda_i^{\text{old}})^2$, which is the direction where the new stimulus most exceeds the perceiver’s prior variance estimate. $\square$

Example 17 (The punchline of a joke). *A joke sets up an expectation (low-variance interpretation) and then delivers a punchline that recontextualizes the setup under a different interpretation (high-variance direction suddenly becomes relevant). The compression progress is large because the entire setup sequence becomes more compressible under the new interpretation.*

Example 18 (The key change in music). *A modulation from the home key to a distant key creates a large projection onto a previously low-variance direction in harmonic space. The compression progress spikes at the moment of modulation and again at the return to the home key (when the listener’s model updates to accommodate the overall tonal plan).*

Example 19 (The “aha” in a mathematical proof). *The critical step in an elegant proof is one that reveals a connection between previously separate structures. In the eigenspace of mathematical knowledge, this step causes a large eigenvalue increase in a direction that was previously near-zero (the connection was unknown), suddenly making a large body of previously complex results compressible.*

VI. UNIFYING CLASSICAL THEORIES

We now show that the framework subsumes three classical theories as special cases.

A. Birkhoff’s Aesthetic Measure (1933)

Birkhoff defined the aesthetic measure as $M = O/C$, where O is “order” (symmetry, regularity) and C is “complexity” (the number of distinct elements).

Proposition 20 (Birkhoff as a special case). *In the compression framework, Birkhoff’s measure corresponds to*

$$M = \frac{O}{C} = \frac{d - \dim(\text{Fix}(G))}{D_{\text{eff}}(z)}, \quad (16)$$

the ratio of compressed-away dimensions (order) to effective stimulus dimensions (complexity). This is a monotone function of $\mathcal{S}$ in the regime where the stimulus is purely symmetric.

The limitation of Birkhoff’s measure is that it treats order and complexity as competing quantities. Our framework shows that they are components of a single functional: the structured surprise multiplies (not divides) complexity by compressibility.

B. Berlyne’s Inverted-U (1971)

Berlyne found that hedonic value follows an inverted-U as a function of stimulus “arousal potential” (complexity, novelty, surprise).

Proposition 21 (Berlyne as a special case). *The structured surprise $\mathcal{S}(x; \Sigma)$ is an inverted-U function of the stimulus’s effective dimensionality $D_{\text{eff}}(z)$ when the total energy $E = \|z\|^2$ is fixed:*

$$\mathcal{S} = D_{\text{eff}}(z) \cdot g(D_{\text{eff}}(z)), \quad (17)$$

where g is a decreasing function (compressibility decreases with dimensionality). The product has a unique interior maximum at $D_{\text{eff}}^* \in (1, d)$.

Proof. Fix $\|z\|^2 = E$. As $D_{\text{eff}}(z)$ increases from 1 toward d , the energy spreads across more dimensions. For fixed total energy spread across more dimensions, the per-dimension energy z_i^2 decreases on average, and the compressibility term $1 - \frac{1}{d} \sum z_i^2 / \lambda_i$ depends on how the energy distribution aligns with $\{\lambda_i\}$.

When $D_{\text{eff}} = 1$, all energy is on one dimension: $\mathcal{S} = 1 \cdot (1 - E/(d\lambda_k))$ for some k —low because the prefactor is minimal.

When $D_{\text{eff}} = d$, energy is uniform: $\mathcal{S} = d \cdot (1 - \frac{E}{d} \sum 1/\lambda_i \cdot 1/d)$ —the compressibility term is small because a uniform spectrum cannot be compressed.

The maximum occurs at intermediate D_{eff} , precisely where complexity and compressibility are jointly maximized. $\square$

C. Reward Prediction Error (Neuroscience)

Neuroimaging studies show that aesthetic pleasure correlates with dopaminergic reward prediction error [?] —the brain’s signal that something was *better than expected*.

Proposition 22 (RPE as compression progress). *Compression progress $\dot{A}_t$ is a form of prediction error: it measures the*

discrepancy between the data's compressibility under the old model (prediction) and the new model (updated reality). Specifically,

$$\dot{A}_t = \sum_{i=1}^d z_i^2 \left(\frac{1}{\lambda_i^{(t-1)}} - \frac{1}{\lambda_i^{(t)}} \right) \geq 0, \quad (18)$$

which is always non-negative (learning only improves compression). The signal is strongest when the model update is large—when the perceiver's understanding of the stimulus undergoes a qualitative change.

VII. THE RIEMANNIAN GEOMETRY OF AESTHETIC PREFERENCE

The space of positive definite matrices $\mathcal{S}_{++}^d$ (the space of possible perceivers) carries a natural Riemannian metric—the Fisher information metric, which coincides with the affine-invariant metric on the positive definite cone.

Definition 23 (The aesthetic manifold). *The aesthetic manifold is $(\mathcal{S}_{++}^d, g_F)$, where*

$$g_F(\dot{\Sigma}_1, \dot{\Sigma}_2) = \text{tr}(\Sigma_1^{-1} \dot{\Sigma}_1 \Sigma_2^{-1} \dot{\Sigma}_2) \quad (19)$$

is the Fisher metric at point Σ .

On this manifold, the geodesic distance between two perceivers Σ_1 and Σ_2 is

$$d_F(\Sigma_1, \Sigma_2) = \left\| \log(\Sigma_1^{-1/2} \Sigma_2 \Sigma_1^{-1/2}) \right\|_F, \quad (20)$$

where $\log$ is the matrix logarithm and $\|\cdot\|_F$ is the Frobenius norm.

Theorem 24 (Cultural distance is geodesic distance). *If two cultural groups A and B have perceptual covariances Σ_A and Σ_B learned from their respective artistic traditions, then the divergence in their aesthetic preferences for a stimulus x is bounded by*

$$|\mathcal{S}(x; \Sigma_A) - \mathcal{S}(x; \Sigma_B)| \leq L \cdot d_F(\Sigma_A, \Sigma_B) \cdot \|x\|^2, \quad (21)$$

where L is a Lipschitz constant depending on the eigenvalue bounds of Σ_A and Σ_B .

Proof. The structured surprise $\mathcal{S}(x; \Sigma)$ depends on Σ through the eigenvalues $\{\lambda_i\}$. By the perturbation theory of eigenvalues (Weyl's inequality), a perturbation $\delta\Sigma$ in the Fisher metric changes each eigenvalue by at most $\|\delta\Sigma\|_F / \|\Sigma^{-1}\|_2$. The result follows from bounding $|\partial\mathcal{S}/\partial\lambda_i|$ and summing over dimensions. $\square$

This theorem formalizes the intuition that cultural aesthetic differences are proportional to the distance between learned perceptual structures, not arbitrary. Two cultures that share similar artistic exposure (small geodesic distance) will agree on aesthetics; two with very different exposure will diverge predictably.

VIII. TESTABLE PREDICTIONS

The framework generates quantitative predictions:

- 1) **The expertise shift.** Experts prefer stimuli with higher effective dimensionality than novices (because D_{eff}^P is larger for experts, shifting the $\mathcal{S}$ -maximum rightward). This is measurable by comparing preferred music complexity in trained musicians vs non-musicians.
- 2) **Cross-domain transfer.** If a person develops expertise in visual art (expanding D_{eff} for visual stimuli), the framework predicts *no* transfer to musical aesthetic preferences (the eigenspaces are domain-specific). But training in abstract pattern recognition *should* transfer partially, because abstract patterns share eigenvectors across modalities.
- 3) **The broken symmetry preference.** Stimuli with approximate symmetry ($\|\epsilon\| / \|x_{\text{sym}}\| \approx 0.05\text{--}0.15$) should be preferred over both perfect symmetry and strong asymmetry. This ratio is computable from image statistics and comparable against preference data.
- 4) **Compression progress predicts chills.** In music, “chills” (frisson) should occur at moments of maximal $\dot{A}_t$: key changes, unexpected harmonic resolutions, the entrance of a new voice. This is testable by computing $\dot{A}$ from MIDI representations and comparing against physiological chill measurements.
- 5) **Mathematical elegance is compressibility.** Among proofs of the same theorem, mathematicians should rate as more “elegant” those with lower Kolmogorov complexity (shorter description in a natural formal system). This is approximately testable using proof length in a fixed proof assistant.

IX. DISCUSSION

The theory proposed here has several limitations that merit discussion.

The linearity assumption. We model perceptual spaces as linear (inner product spaces with PCA-like structure). Real perceptual spaces are likely nonlinear manifolds. Extending the framework to kernel PCA or autoencoders would capture nonlinear structure at the cost of analytical tractability.

The Gaussian assumption. The aesthetic functional is cleanest when stimuli are Gaussian-distributed. Real distributions are heavy-tailed and multimodal. The structured surprise definition is more general, but the closed-form results in Theorems ?? and ?? rely on Gaussian or near-Gaussian behavior.

The compression-beauty gap. Not everything that is compressible is beautiful (a repeating tile pattern is highly compressible but not particularly beautiful after the first glance). The framework addresses this through the effective dimensionality factor, but a complete theory may need additional mechanisms (attentional dynamics, social context, personal association).

The measurement problem. The perceptual covariance Σ_P is not directly observable. It must be estimated from behavioral data (preference rankings, similarity judgments) or neural data (representational similarity analysis). The theory's predictions are therefore conditional on accurate estimation of the perceiver's eigenstructure.

X. CONCLUSION

We have developed a mathematical theory of aesthetic experience based on three rigorous results:

- 1) The *aesthetic optimum* (Theorem ??): the most beautiful stimulus has a power spectrum that follows a power law of the perceiver's eigenvalues, $z_i^2 \propto \lambda_i^\alpha$ with $\alpha \in (0, 1)$, lying between perfect predictability and randomness.
- 2) *Symmetry as compression* (Theorem ??): symmetry under a group G compresses description length by a factor of $|G|$, and broken symmetry adds information on top of this scaffold.
- 3) *Compression progress* (Theorem ??): temporal aesthetic experience (music, narrative, proof) is driven by model updates that increase compressibility, with the strongest signal from updates that affect previously low-variance directions.

These results unify Birkhoff's aesthetic measure, Berlyne's inverted-U, and reward prediction error theory as special cases, and generate five quantitative predictions testable against existing data.

The deepest implication connects to the broader Geometric Series. In PCA-Matryoshka [?], we showed that eigenvalues are importance weights for embedding compression: truncating in the PCA basis preserves meaning while truncating in a random basis destroys it. The aesthetic theory developed here suggests that this is not merely a useful engineering heuristic—it reflects something about how perception works. Beauty may be the subjective experience of encountering structure that compresses well in the basis evolution gave us to understand the world.

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